Assignment 9: Unsupervised Learning

Introduction

The aim of this assignment is to use unsupervised learning to analyze the Wine dataset. The dataset contains different chemical properties of wine samples. Clustering techniques were used to group similar wines based on their features. Principal Component Analysis (PCA) was also used to reduce the data into two dimensions, making the clusters easier to visualize and understand.

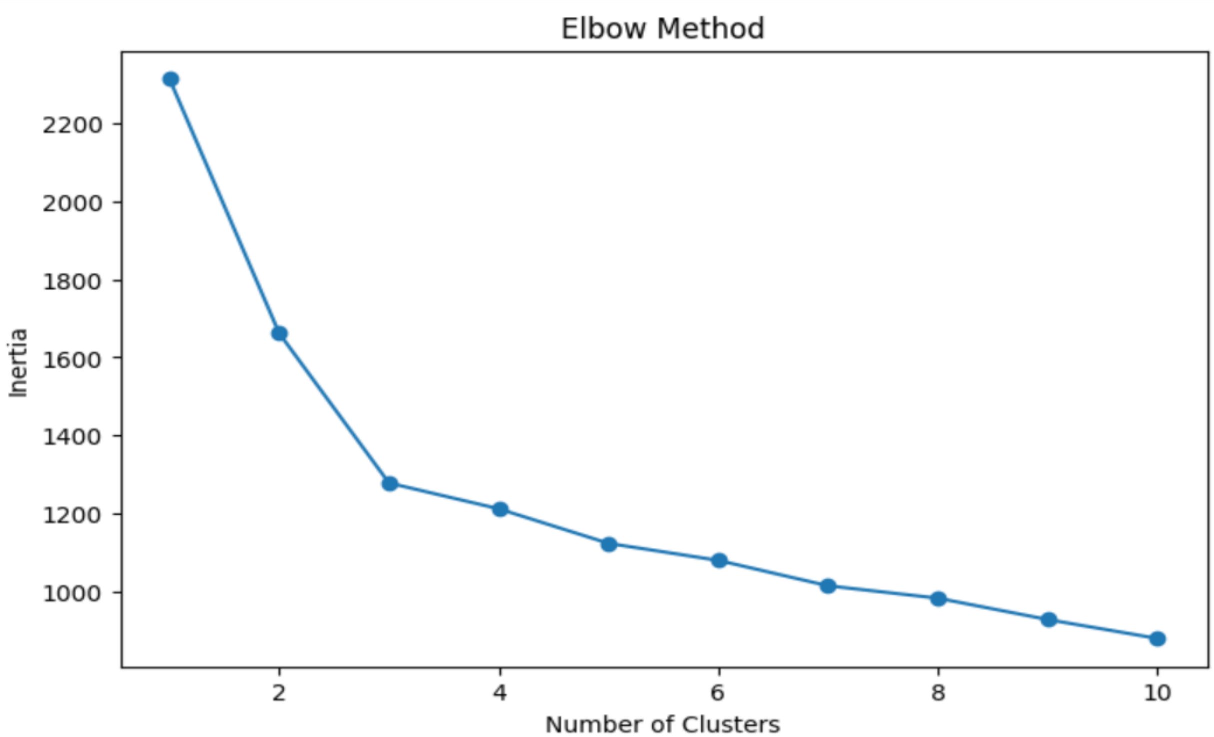
Data Preprocessing

The Wine dataset was first examined for missing values, and no missing data was found. Since clustering algorithms are sensitive to differences in feature scales, all numerical features were standardized using the StandardScaler. This ensured that each feature contributed equally during clustering.

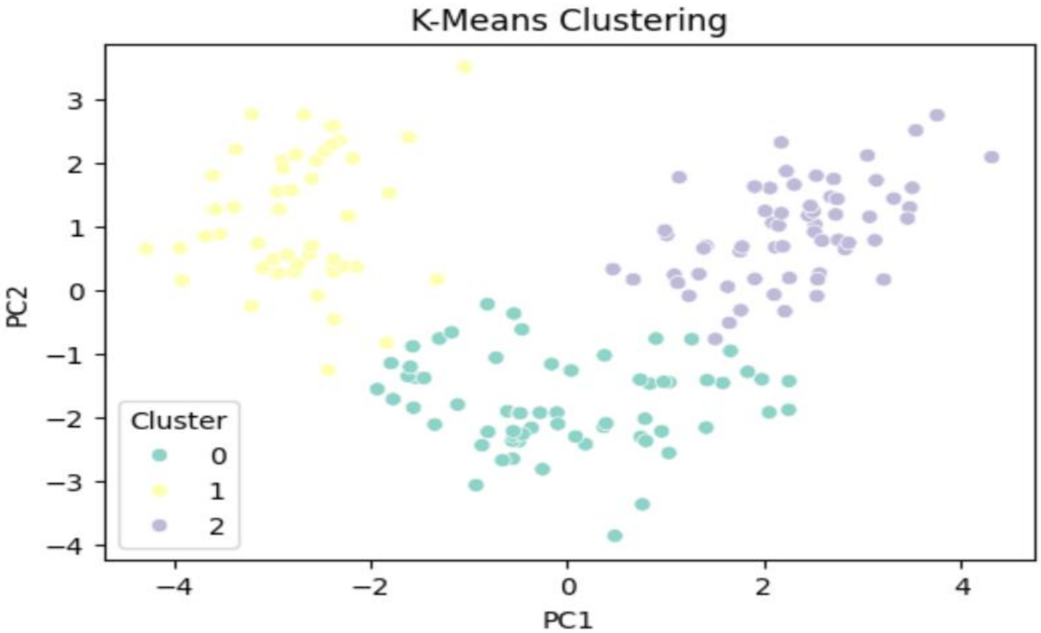
Clustering Methodology

K-Means Clustering

K-Means clustering results after reducing the dataset to two principal components (PC1 and PC2) using PCA. The data points are grouped into three distinct clusters (0, 1, and 2), represented by different colors. Most clusters are well separated, indicating that the K-Means algorithm successfully identified groups of wine samples with similar characteristics. This visualization confirms that K = 3 is an appropriate choice for clustering the Wine dataset.**Elbow Method**

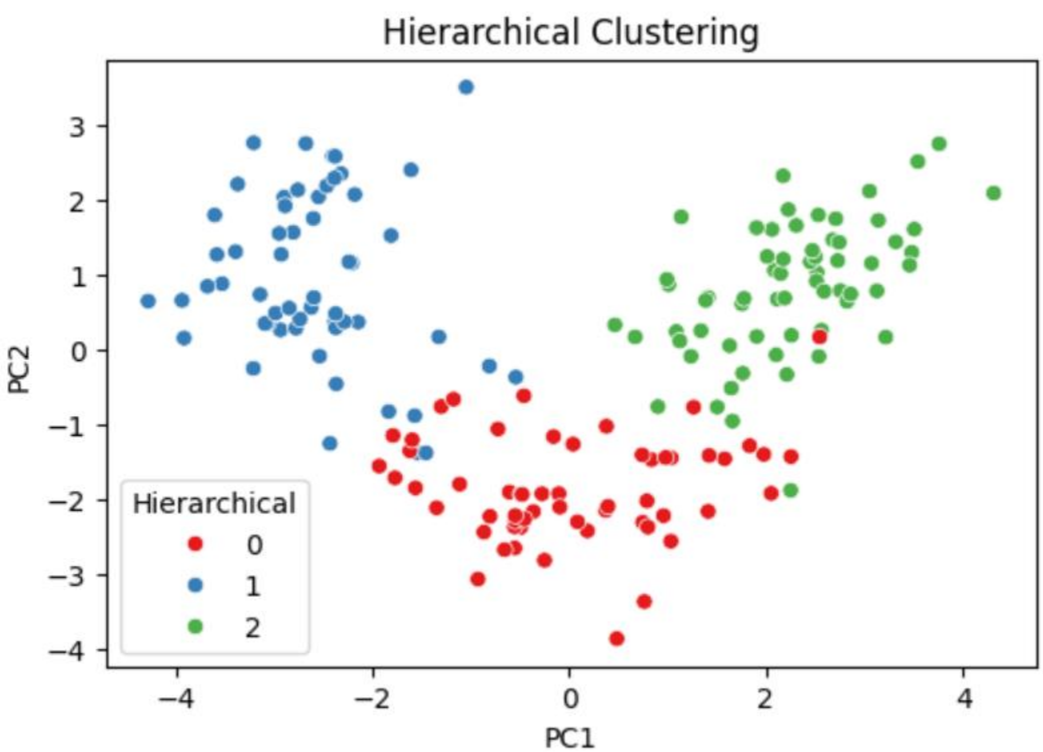


K-Means Clustering



Hierarchical Clustering

The Hierarchical Clustering model achieved a Silhouette Score of 0.2774, which is close to 0.28 obtained by the K-Means model. This indicates that the clusters are reasonably well separated, although there is some overlap between them. The positive silhouette score shows that the model has successfully grouped similar wine samples together.



PCA

Principal Component Analysis (PCA) was used to reduce the original features into two principal components. This made it easier to visualize the clustering results while preserving most of the important information in the dataset. PCA clearly displayed the separation between the clusters created by both clustering algorithms.

Model Evaluation

The clustering models were evaluated using the Silhouette Score.

- **K-Means Silhouette Score:** 0.2849
- **Hierarchical Clustering Silhouette Score:** 0.2774

The silhouette scores of the two clustering models were compared to evaluate their performance. K-Means achieved a silhouette score of 0.2849, while Hierarchical Clustering achieved 0.2774. Since the K-Means score is slightly higher, it indicates that its clusters are better separated and more compact. Therefore, K-Means is selected as the better-performing clustering algorithm for the Wine dataset, although both models produced satisfactory clustering results.

Deployment Strategy

Some challenges that may occur include:

- The model may take longer to process if a large amount of wine data is added.
- It may become difficult to handle an increasing number of wine samples over time.
- New wine samples may have different characteristics, which can reduce the accuracy of the clusters.
- The model should be checked and updated regularly to maintain good performance.

Monitoring Strategy

The deployed model should be monitored by:

- Checking the silhouette score regularly to make sure the clustering performance remains good.
- Watching for changes in the wine data that could affect the clustering results.
- Retraining the model from time to time using new wine samples.
- Keeping a record of the clustering results to ensure the model is working correctly.
- Reviewing the number of clusters and updating it if new data changes the grouping pattern.

Conclusion

The Wine dataset was analyzed using unsupervised learning techniques. Both K-Means and Hierarchical Clustering were used to group similar wine samples, and PCA was used to reduce the data into two dimensions for easier visualization. Based on the silhouette scores, K-Means performed slightly better than Hierarchical Clustering. Finally, simple deployment and monitoring strategies were suggested to help keep the model accurate and reliable when used with new data.